Mathematics Revision Test 33

Another speed test. 20 questions this week

Get your watch out for this exercise and get someone to time you please.

1. Find the PRODUCT of 9 and 7:

_____

2. (0 x 7) + (7+ 0) = _____

3. From noon till 12:45pm = _____minutes

4. $\frac{1}{5}$ of 400 = _____

5.

Show the ACUTE angle on the quadrilateral above

6.

Now show the OBTUSE angle on the quadrilateral above

7. $\frac{3}{5}$ of 400 = _____

8. Add 6 to the PRODUCT of 7 and 7.

_____

9. How much greater than is (3 x 10) than (4 x 7)? _____

10. 2 Km 90 m = _____ m

11. $\frac{1}{2}$ Km – 397m = _____

12. £0.15 x 8 = £ _____

13. 16p x 7 = £_____

14. £0.23 + £1.77 = £_____

15. £0.64 ÷ 8 = £_____

16. 20 31 54 69

Add the odd numbers = _____

17. Find the total of 24 and 79 =

_____

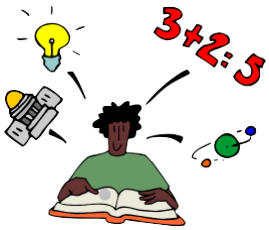
18. 16 ÷ 10 = _____

19. $\frac{7}{10}$ of 1 Kg = _____

20. Add one-sixth of 54 to one-third of 27 = _____

Seconds taken: _____

Marks this week: _____



Calculation Ladder

Ok, heads down and concentrate.

Think in terms of pounds as you work it out e.g. 2.5 = £2.50

Ladder 11

1.	£66
2.	deduct £11
3.	find 10%
4.	double it
5.	add £14
6.	10% of this
7.	Multiply by 6
8.	50% of this
9.	10% of this
10.	Double it
	Your answer is

Ladder 12

1.	£45
2.	find 20%
3.	Add £12
4.	find 50%
5.	Multiply by 10
6.	Add £5
7.	10% of this
8.	x 8
9.	find 50%
10.	10% of this
	Your answer is

Are you enjoying these?

How To Add Fractions

You cannot add together fractions unless they have the same **denominator** (bottom number).
When you have made both fractions have the same denominator, you simply add them together.
Example: To add $\frac{1}{3}$ to $\frac{1}{9}$ you must give them the same denominator. $\frac{1}{3}$ is the same as $\frac{3}{9}$.

So: $\frac{3}{9} + \frac{1}{9} = \frac{4}{9}$ The same applies to subtracting fractions

Now try these. (The first one has been done for you.)

$\frac{6}{8} + 1\frac{1}{4}$	2	$\frac{1}{5} + \frac{2}{20}$	
$\frac{1}{4} + \frac{1}{6}$		$\frac{5}{14} + \frac{1}{2}$	
$\frac{3}{4} + \frac{1}{16}$		$\frac{1}{3} + \frac{5}{15}$	
$\frac{3}{5} + \frac{4}{10}$		$\frac{2}{10} + \frac{6}{30}$	
$\frac{5}{6} + \frac{1}{2}$		$\frac{1}{3} + \frac{1}{6}$	

How To Add Fractions

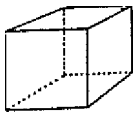
$\frac{1}{3} + \frac{2}{18}$		$\frac{1}{3} + \frac{2}{21}$	
$\frac{2}{5} + \frac{1}{2}$		$\frac{1}{2} + \frac{1}{3}$	
$\frac{1}{4} + \frac{1}{3}$		$\frac{2}{3} + \frac{5}{6}$	
$\frac{6}{12} + \frac{6}{8}$		$\frac{1}{3} + \frac{1}{24}$	
$\frac{1}{2} + \frac{3}{10}$		$\frac{1}{2} + \frac{1}{22}$	
$\frac{5}{8} + \frac{2}{16}$		$\frac{1}{6} + \frac{1}{12}$	

Solid shapes and their nets

Read through this carefully then complete the next sheet.

1. The cube

   All its six faces and *all* it's angles are right angles

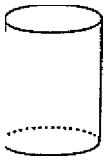


2. The cuboid

   All its faces are rectangles (so a cube is a special kind of cuboid). It tessellates - common examples of such tessellations are brick walls, rows of books and paving stones.

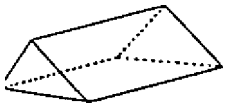
3. The cylinder

   Two identical circular faces andone curved surface whose net is a rectangle. The usual shape for metal packages {i.e. tin cans). whereas cardboard packages are usually cuboids



4. Triangular prism

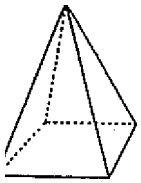
   A cotnmon tent shape. Two identical triangles and three rectangular faces



**All the above four solids can be called prisms. The prism must have two identical parallel faces at either end and its cross-section, if sliced parallel to these faces, must also be identical.

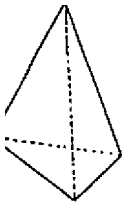
5. Square pyramid

   The base is *a* square and the other faces are identical triangles, as with the Egyptian pyramids.



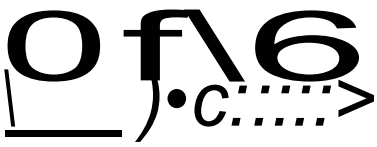
6. Tetrahedron

   Four triangular faces. It could be thought of as a pyramid on a triangular base. A regular tetrahedron has equilateral triangles as faces.



7. Cone

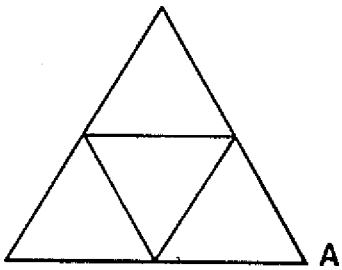
   This can be seen as a pyramid on a circular base. The net is a circle (the base) and the sector of a circle. The greater the sector, the flatter the angle of the cone.

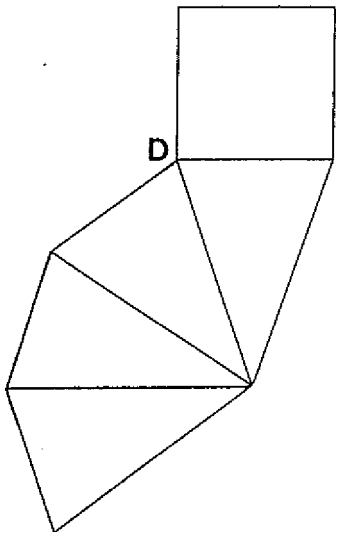


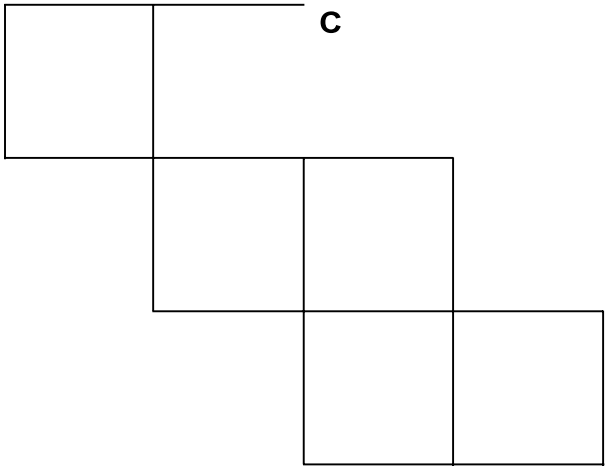
Solid shapes and their nets

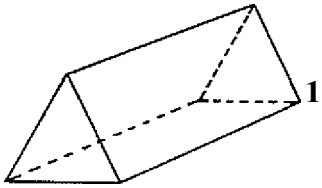
Here are the names of 7 shapes:	Number	Net
Cube		
Cuboid		
Cylinder		
Cone		
Triangular prism		
Pyramid		
Tetrahedron		

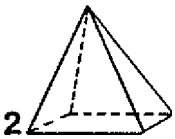
Look below at the shapes (they are numbered) and the nets (lettered), and match them with the right name. Answers overleaf.

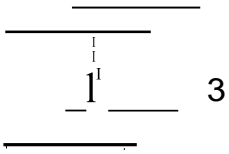


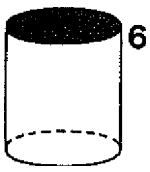


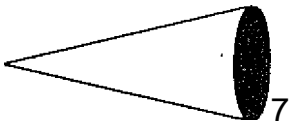


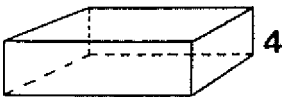


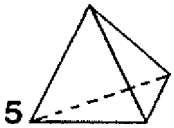


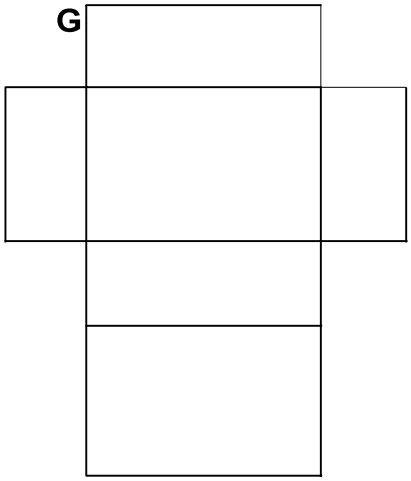


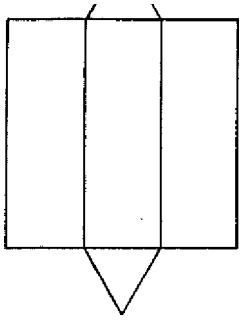












Write the missing sign in each box below to make the sum correct:

1. 3 + 24 + 21 = 4 ☐ 12

2. 28 ☐ 0 = 28

3. 6 ☐ 12 = $\frac{1}{2}$

4. 31 = (17 ☐ 2) - 3

5. 0.8 + 0.34 = 1 ☐ 0.1 + 0.04

6. £2.14 ☐ £1.11 + 90p = £2 - 7p

In these sums the SAME sign must be written in both boxes to make them correct:

7. 15 ☐ 2 = 5 ☐ 6

8. 7 + 12 ☐ 2 = 37 ☐ 20

9. 41 = 58 ☐ 1 ☐ 17 + 1

10. 21 ☐ 3 = 56 ☐ 8

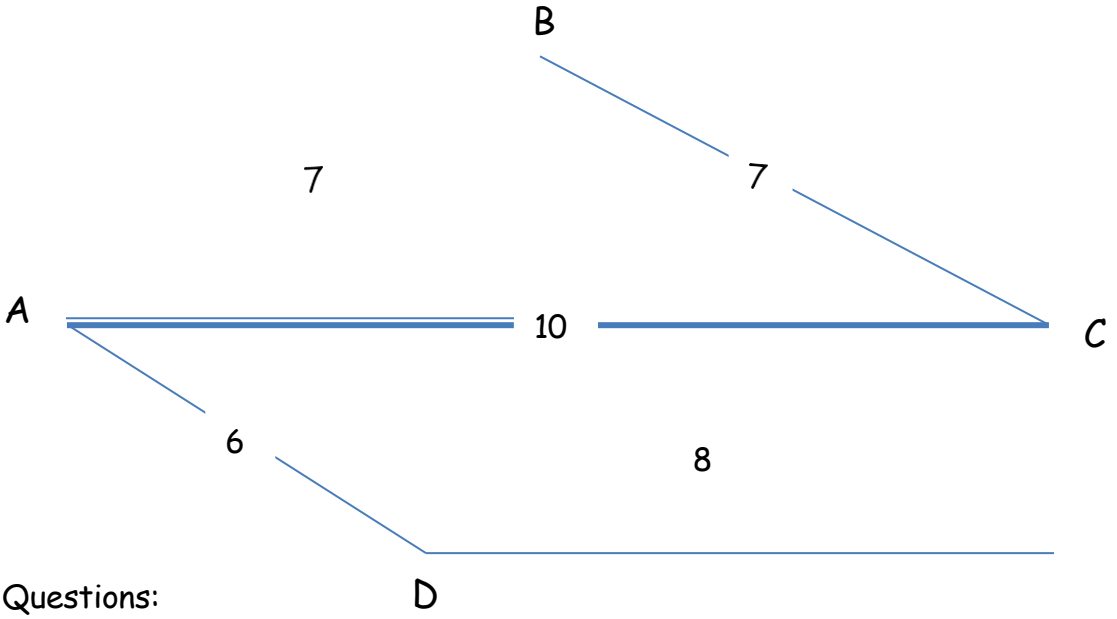
Change the order of the figures in these numbers to make the largest number possible:

1. 10571 _____ 12. 23723 _____

13. 90491 _____ 14. 6859 _____

15. 48670 _____ 16. 132413 _____

This simple map shows roads between villages and the distances, in miles:



Questions:

17. Tom lives 4 miles from D, on the road to A. Some days he walks to school in A. If he walks at 3 miles an hour how long does the walk take him?

_____

18. Alfie cycles the full circuit from A to B to C to D and back to A. It takes him 2 hours non-stop. What is his average speed in miles per hour?

_____

19. A lorry driver takes a load from A to C along the shortest route, averaging 30 mph. How long does the journey take him?

_____

20. Miss Pendleton cycles from A to B, where she does some shopping. She shops for 30 minutes and then returns to A. She left A at 10.00 am and her cycling speed is 7 miles an hour. What time does she get back to A?

_____

21. Sam's car has an average fuel consumption of 12 miles per litre. He drives from A to C to D and back to A. If petrol costs £1.37 a litre, how much does this journey cost him?

_____

22. Charlie, preparing for the marathon, does a training run around the full circuit A-B-C-D-A. It takes him 4 hours to complete the run, running at a consistent pace. How long did the leg from B to C take him?

_____

Complete these sums:

23. $9\frac{11}{22} + 4\frac{3}{4} =$ _____

24. $12\frac{1}{4} - 6 =$ _____

25. $2\frac{2}{3} + 3\frac{1}{3} + 14 =$ _____

26. $4\frac{11}{55} + \frac{44}{55} =$ _____

Place a tick in the box if a statement is true and x if it is not correct:

27. Half of three dozen is sixteen. ☐

28. 3 kilos of apples at 85p a kilo will cost under £2.60. ☐

29. There are 28 days in two fortnights. ☐

30. Three quarters of 16 is twelve. ☐

31. Not all triangles have three sides. ☐

32. Three sixths is equal to five tenths. ☐

33. There are 100g in a kilo. ☐

34. 115 minutes is the same as one hour and three quarters. ☐

35. A square can sometimes have unequal sides. ☐

36. Most months have 30 days ☐

37. 20% of 400 is 80 ☐

Now do these sums:

38.
```
   845
   562
+  997
  ____
```

39.
```
   855
    13
+  1365
  _____
```

40.
```
   591
-  399
  ____
```

41.
```
   803
-   78
  ____
```

42.
```
   743
×    5
  ____
```

43.
```
   811
×   12
  ____
```

Time questions:

44. How would a quarter to six in the morning be shown on a digital clock? _____

45. How long is the time between 19.54 and 20.12? _____

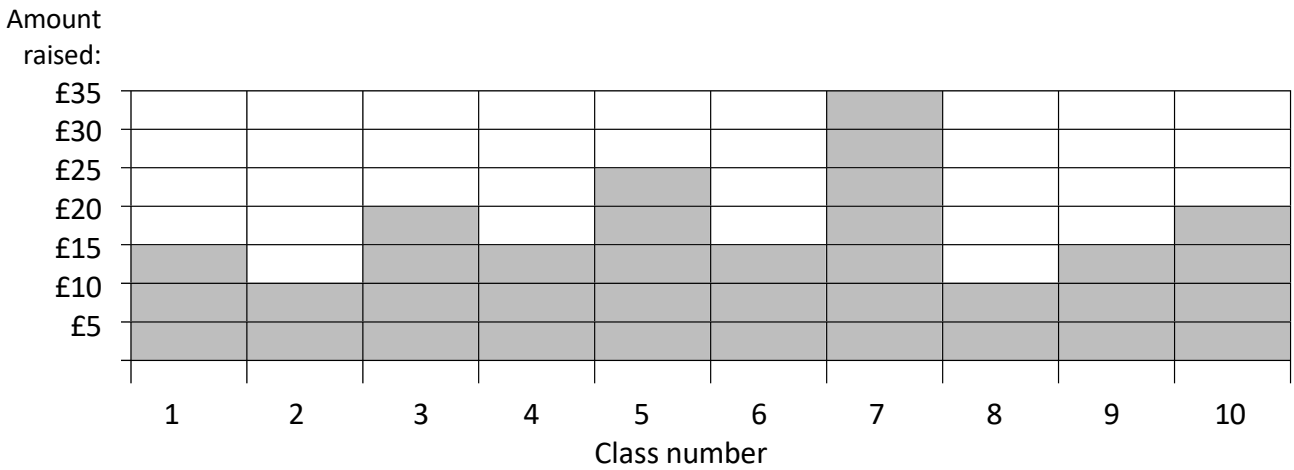
46. The train arrived exactly 27 minutes late, at 10.15am. When should it have arrived? _____

47. How long between half past eight in the evening and 1.20am? _____

48. Lesson 1 starts at 9.00am. Then there is a 20 min break. Lesson 2 ends at 11.00am. Each lesson is the same length. How long is a lesson? _____

49. The flight from London to Inverness takes one hour thirty five minutes. You take off at 11:35am. When do you land? _____

Penny for the Guy raised money towards the cost of a new school swimming pool. Each class raised the amount shown in this block graph.



Look at the graph carefully and answer these questions:

50. How much was raised by the ten classes? _____

51. Which two classes raised the most? _____

52. What fraction of the total did these two classes raise? _____

53. Which four classes collected the same amount? _____

54. What was the average amount collected per class? _____

55. How many classes collected less than the average? _____

56. How many classes collected more than the average? _____

57. What percentage of the grand total was the total sum raised by classes 2, 3, 5, and 7? _____

(Use this space for any working out)

In the following questions multiply each number by 2 and add 9. The first is done for you.

58. 8 → 25 11 → ☐ 24 → ☐

59. 200 → ☐ 42 → ☐ 91 → ☐

Now divide by 3 and subtract 8.

60. 54 → ☐ 111 → ☐ 75 → ☐

END OF TEST

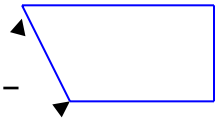
If you have time

- **Go back and complete any questions you missed out.**
- **Check your answers carefully.**

ANSWERS - COMPLETE MATHS – PART 35

Maths
Revision Sheet

1. 63
2. 7
3. 45
4. 80
5. acute –
6. obtuse —
7. 240
8. 55
9. 2
10. 2090
11. 103m
12. £1.20
13. £1.12
14. £2
15. £0.08
16. 100
17. 103
18. 1.6
19. 700
20. 18



Calculation Ladders
Ladder 1 = £1.50
Ladder 2 = £4.40

How To Add Fractions
Sheet 1

2	3/10
5/12	6/7
13/16	2/3
1	2/5
1 1/3	1/2

Sheet 2

4/9	3/7
9/10	5/6
7/12	1 1/2
1 1/4	3/8
4/5	6/11

3/4 $\frac{1}{4}$

Shapes and their nets

Shape	Number	Net
Cube	3	C
Cuboid	4	G
Cylinder	6	B
Cone	7	E
Triangular Prism	1	F
Pyramid	2	D
Tetrahedron	5	A

Maths Type 3 Paper 1

1. x
2. +
3. ÷
4. x
5. +
6. -
4. x
8. -
9. -
10. ÷
11. 75110
12. 73322
13. 99410
14. 9865
15. 87640
16. 433211
17. 40 minutes
18. 14 mph
19. 20 minutes
20. 12.30 pm
21. £2.74
22. 1 hour
23. $14\frac{1}{4}$
24. $6\frac{1}{4}$
25. 20

ANSWERS - COMPLETE MATHS – PART 35

26. 5
27. x
28. ✓
29. ✓
30. ✓
31. x
32. ✓
33. x
34. x
35. x
36. x
37. ✓
38. 2404
39. 2233
40. 192
41. 725
42. 3715
43. 9732
44. 05:45
45. 18 minutes
46. 9.48 am
47. 4 hours 50 mins
48. 50 mins
49. 1.10 pm
50. £180
51. 5 and 7
52. 1/3
53. 1,4,6 and 9
54. £18
55. 6
56. 4
57. 50%
58. 25, 31, 57
59. 409, 93, 191
60. 10, 29, 17